

上海交通大学试卷 (B 卷)

(2013 至 2014 学年 第 1 学期)

班级号 _____ 学号 _____ 姓名 _____

课程名称 _____ 数字电子技术 (期末考试) _____ 成绩 _____

PART I: Multiple Choice (Choose the one alternative that best completes the statement or answers the question.) [20']

1	2	3	4	5	6	7	8	9	10

1. A flash ADC uses _____ []

- A. counters
- B. flip-flops
- C. an integrator
- D. op-amps

2. Asynchronous counters are often called _____ counters. []

- A. binary
- B. toggle
- C. ripple
- D. flip-flop

3. Which type of gate can be used to add two bits? []

- A. NAND
- B. NOR
- C. XOR
- D. XNAND

4. The resolution of a 12-bit DAC is nearly _____ []

- A. 0.1%
- B. 0.01%
- C. 0.25%
- D. 0.025%

5. Like the latch, the flip-flop belongs to a category of logic circuits known as _____ []

- A. monostable multivibrators
- B. bistable multivibrators
- C. astable multivibrators
- D. one-shots

6. A feature that distinguishes the J-K flip-flop from the S-R flip-flop is the _____ []

- A. toggle condition
- B. preset input
- C. type of clock
- D. clear input

我承诺，我将严
格遵守考试纪律。

承诺人：_____

题号	PART I	PART II						
		1	2	3	4	5	6	
得分								
批阅人(流水阅 卷教师签名处)								

7. Which of the following is correct for full-adders? []

- A. Full-adders have the capability of directly adding decimal numbers.
- B. Full-adders are used to make half-adders.
- C. Full-adders are limited to two inputs, since there are only two binary digits.
- D. In a parallel full-adder, the first stage may be a half-adder.

8. The modulus of a counter is _____ []

- A. the number of flip-flops
- B. the actual number of states in its sequence
- C. the number of times it recycles in a second
- D. the maximum possible number of states

9. Which one of the following is an example of a counter with a truncated modulus. []

- A. Modulus 8
- B. Modulus 14
- C. Modulus 16
- D. Modulus 32

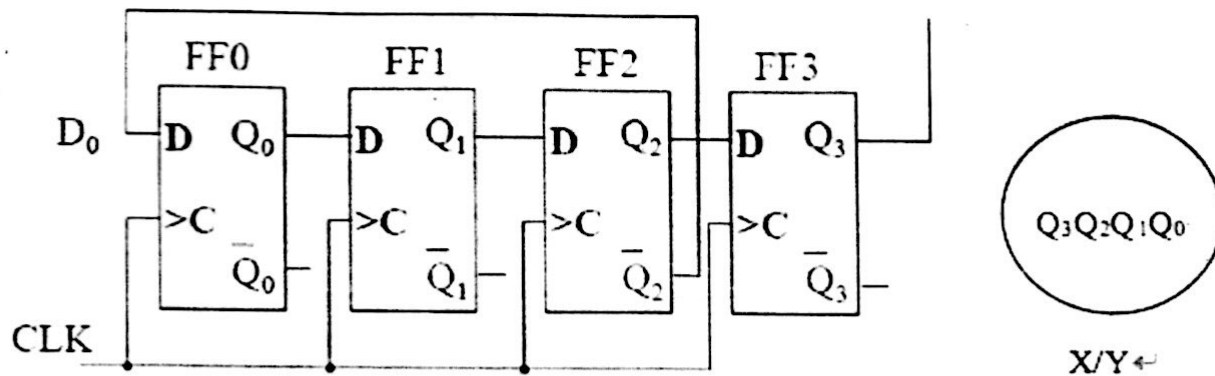
10. A modulus-10 Johnson counter requires _____ []

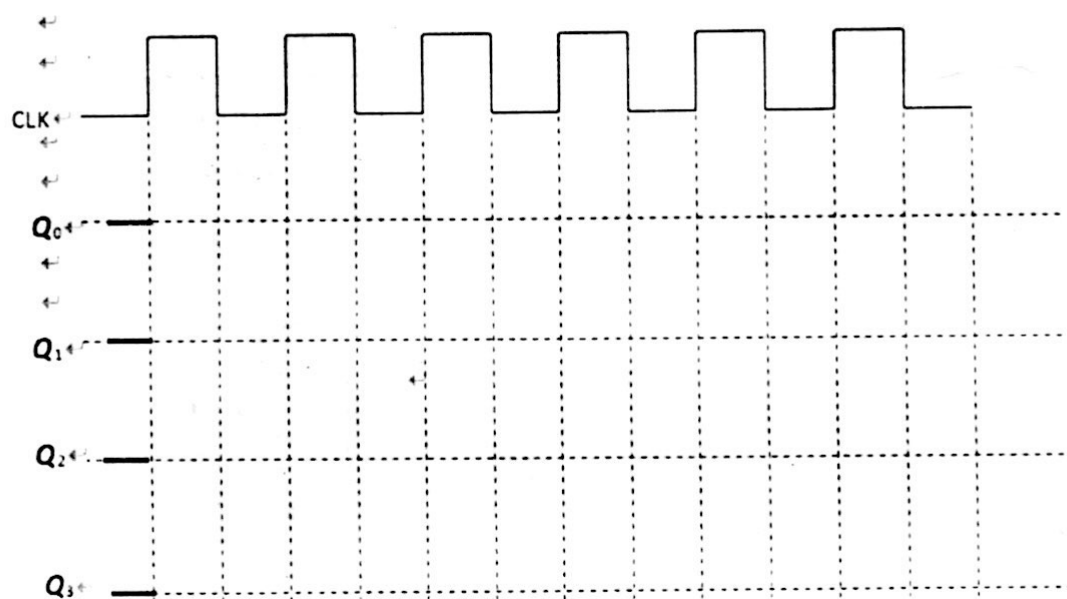
- A. twelve flip-flops
- B. ten flip-flops
- C. five flip-flops
- D. four flip-flops

PART II: Problems

1. Analyze a digital logic circuit shown in the following logic diagram. [15']

- (1) Draw the state diagram of the circuit.
- (2) Draw the waveforms of the outputs (initial state is 0000).
- (3) Describe the function of the circuit.



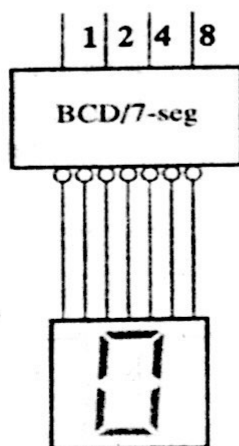
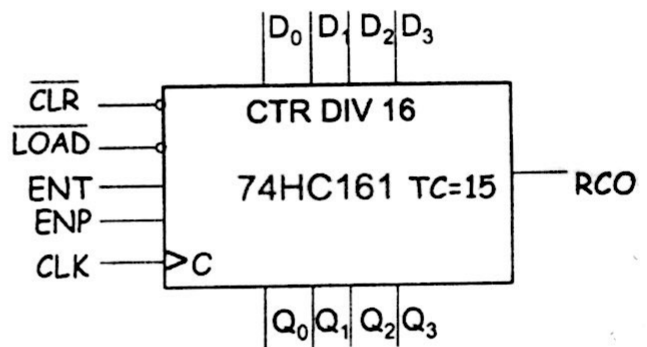
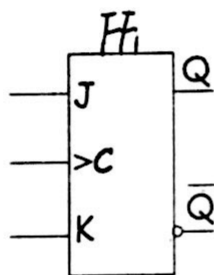
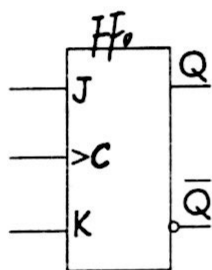


Figures for problem 1

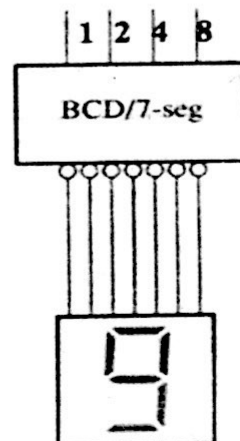
2. Design a sequential circuit for a 24-hour digital clock with a 4-bit binary counter 74HC161 which showed as follows and J-K flip-flops, meanwhile the sequential circuit only displays hours with BCD/7-seg and LED, the sequence should be displayed as 01, 02, 03, ... , 22, 23, 24, 01, 02, ... [10']

74HC161 Function Table

CLK	CLR	LOAD	ENP	ENT	Function
↑	0	X	X	X	Clear
↑	1	0	X	X	Preset
X	1	1	0	1	Inhibit
X	1	1	X	0	Inhibit (RCO=0)
↑	1	1	1	1	Count



Tens-Digit



Unit-Digit

Figures for problem 2

3. Design a sequential circuit with sequence shown in the following state diagram with **J-K Flip-flops** and **logic gates**. Please describe your design procedure and derive the logic expressions of all J-K Flip-flop inputs. (Initial state 000) [15']

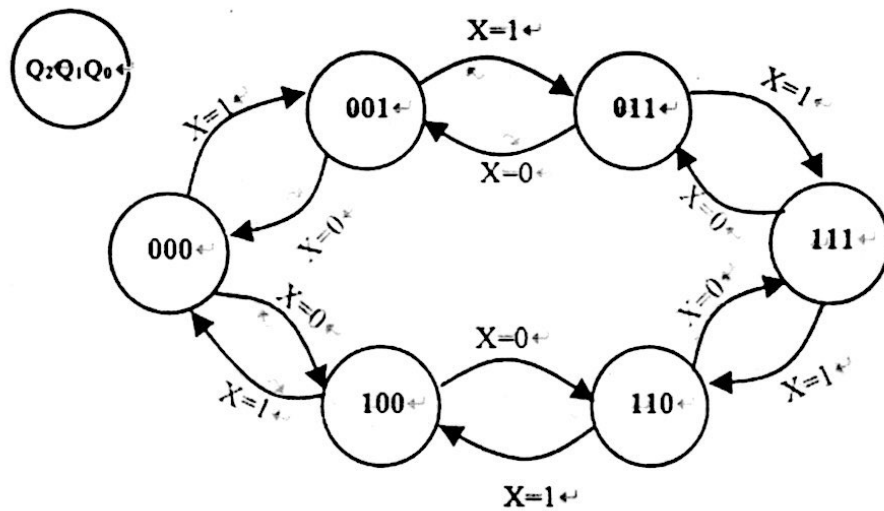
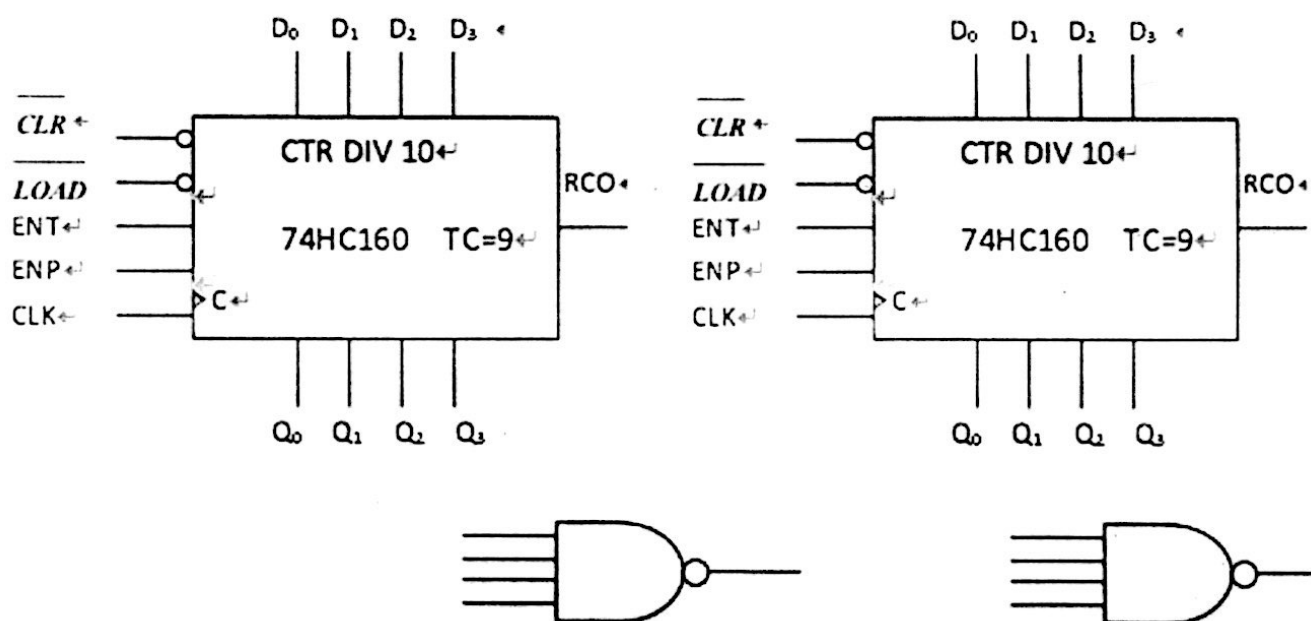


Figure for problem 3

4. Design a **modulus-43** counter with decimal counters **74HC160** and **TWO NAND** gates.
The sequence should be 0, 1, 2, ..., 11, 68, 69, ..., 98, 0, 1, [16']

74HC160 Function Table

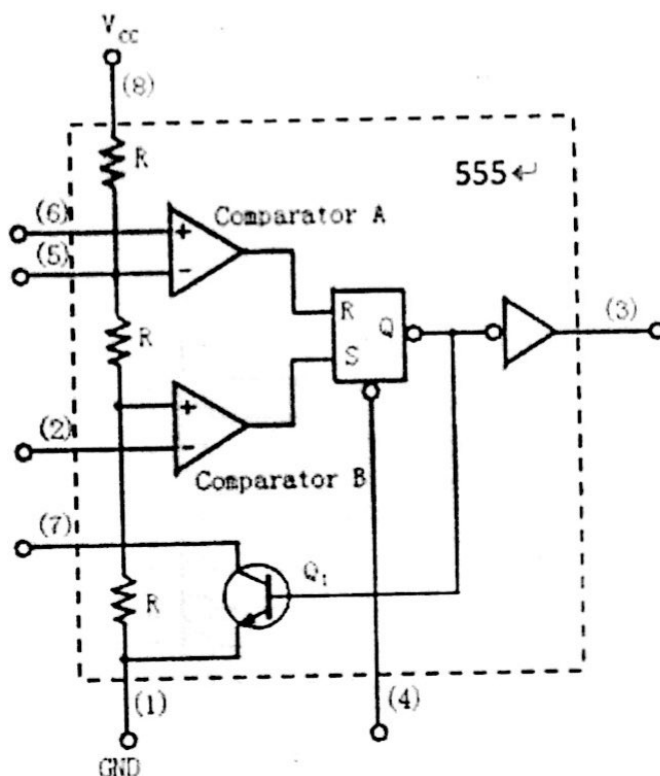
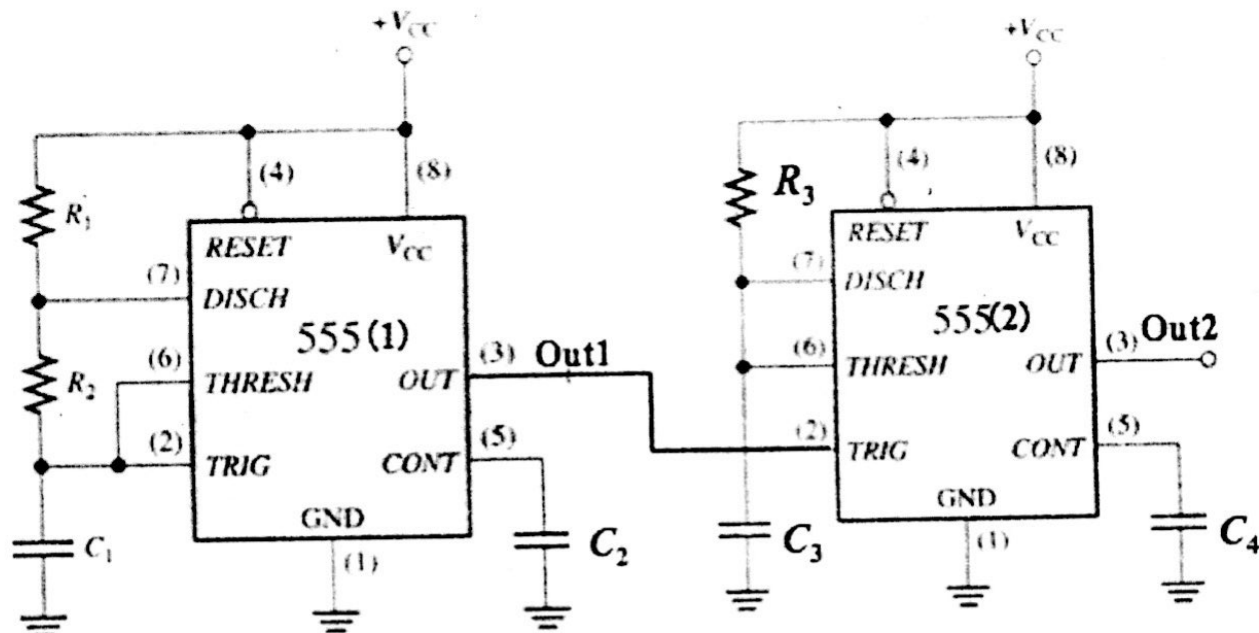
CLK	$\overline{CLR}$	$\overline{LOAD}$	ENP	ENT	Function
X	0	X	X	X	Clear
$\uparrow$	1	0	X	X	Preset
X	1	1	0	1	Inhibit
X	1	1	X	0	Inhibit (rco=0)
$\uparrow$	1	1	1	1	Count

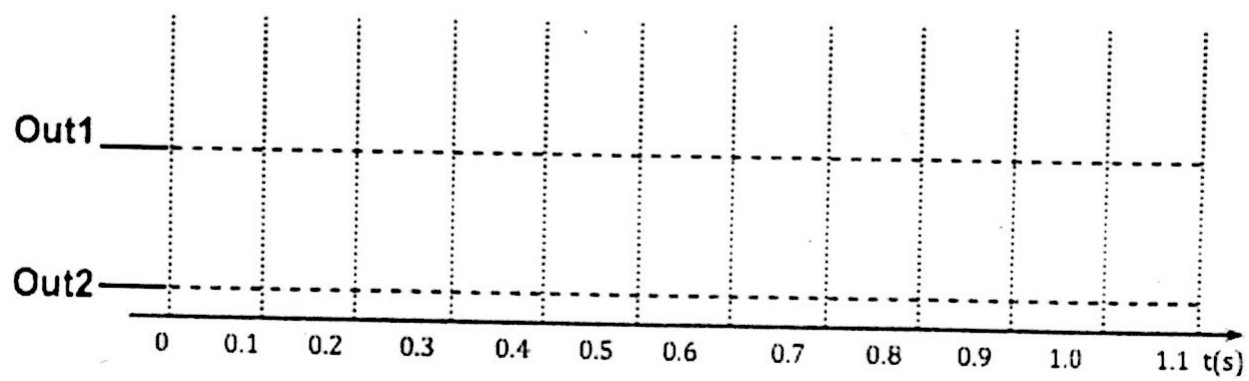


Figures for problem 4

5. For the following circuit, please answer the following questions. ($R_1=100\text{K}\Omega$, $R_2=22\text{K}\Omega$, $C_1=2\mu\text{F}$, $C_2=0.01\mu\text{F}$, $R_3=50\text{K}\Omega$, $C_3=2.2\mu\text{F}$, $C_4=0.01\mu\text{F}$) [12']

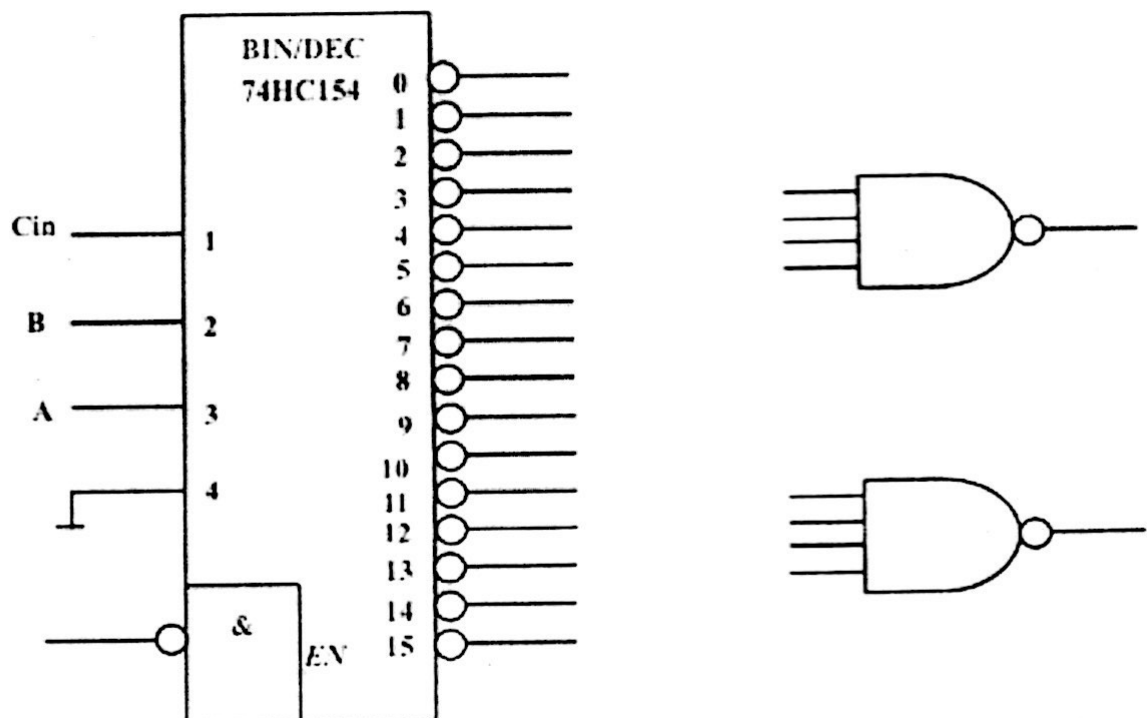
- (1) Describe the operation of the two 555 timers.
- (2) Calculate the period and duty cycle of Out1 if its waveform is periodic.
- (3) Calculate the period and duty cycle of Out2 if its waveform is periodic.
- (4) Draw the waveform of Out1 and Out2 respectively (initial state are Low level).





Figures for problem 5

6. Design a one-bit binary full subtractor with one **1-of-16 decoder 74HC154** and two **NAND** gates. Note that the full-subtractor has **three inputs** including minuend B, subtrahend A, and input borrow Cin and **two outputs** S and Cout. Please describe the design procedure and draw the logic diagram. [12']



Figures for problem 6

上海交通大学试卷 (B 卷)

(2012 至 2013 学年第 1 学期)

班级号 _____ 学号 _____ 姓名 _____

课程名称 数字电子技术 (期末) 成绩 _____

PART I : True/False Write 'T' if the statement is true and 'F' if the statement is false.

1. A D flip-flop is constructed by connecting an inverter between the Set and Clock terminals of an S-R flip-flop. []
2. The 555 timer has three basic operating modes: monostable, bistable, and astable. []
3. An eight-bit D/A converter has a resolution of 0.125. []
4. The term synchronous as applied to counter operations means that the counter is clocked so that each flip-flop in the counter is triggered at the same time. []
5. Once an up/down counter begins its count sequence, it cannot be reversed. []
6. In many cases counters must be strobed in order to eliminate glitches. []
7. A J-K flip-flop can be used as a divide-by-two frequency divider with an output duty cycle of 50%. []
8. A serial-in serial-out shift register transfers data from one line of a parallel bus to another line one bit at a time. []
9. Most sequential circuits contain a combinational logic section and a memory section. []
10. An effective time delay device can be constructed by using the propagation delay characteristic of parallel shift registers. []

Part II: Multiple Choices (Choose the one alternative that best completes the statement or answers the question.)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

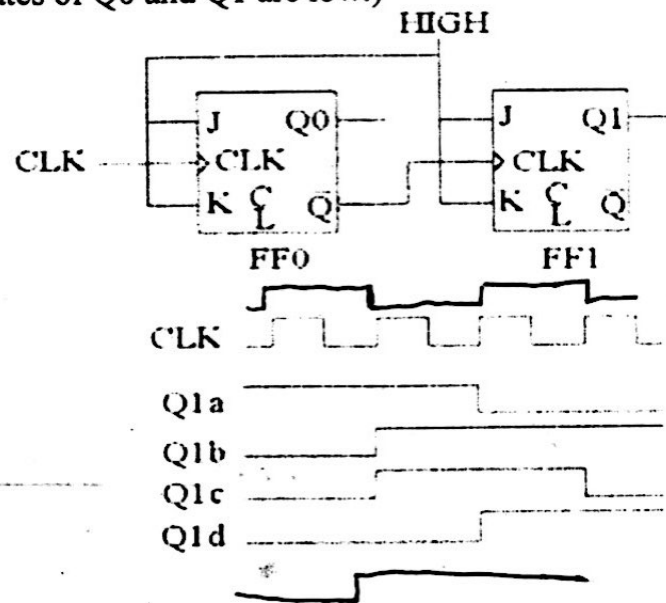
我承诺，我将
严格遵守考试纪
律。

承诺人：

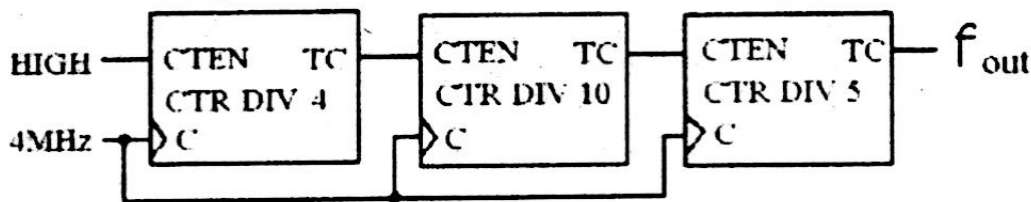
题号	Part I	Part II	Part III				
			1	2	3	4	5
得分							
批阅人(流水阅卷教师签名处)							

- Which of the following is correct for a gated D latch?
 - Q output follows the input D when the ENABLE is high.
 - The output complement follows the input when enabled.
 - The output toggles if one of the inputs is held high.
 - Only one of the inputs can be high at a time.
- Edge-triggered flip-flops must have _____.
 - very fast response times
 - a pulse transition detector
 - at least two inputs to handle rising and falling edges
 - active-low inputs and complemented outputs
- A retriggerable one-shot has a pulse width of 10ms. 3ms after being triggered, another trigger pulse is applied. The resulting output pulse will be _____ ms.
 - 7
 - 3
 - 13
 - 10
- Which of the following is not generally associated with flip-flops?
 - Interval time
 - Set-up time
 - Hold time
 - Propagation delay time

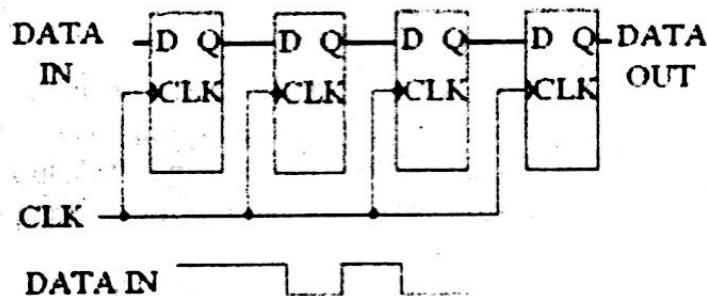
5. The counter shown below is a(n) _____ counter, and the correct output waveform for Q1 is _____. (The initial states of Q0 and Q1 are low.)



- a. synchronous, Q1a
b. asynchronous, Q1b
c. asynchronous, Q1c
d. synchronous, Q1d
6. What is the output frequency of the counter in Figure shown below?

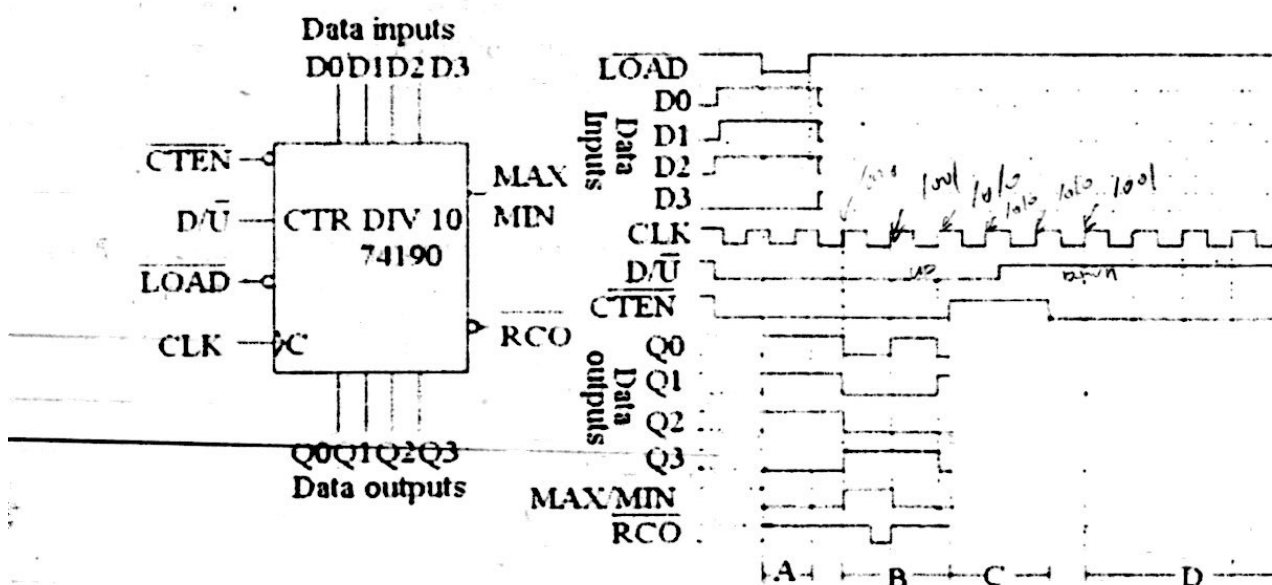


- a. 210.5 kHz
b. 20 kHz
c. 800 Hz
d. 4 MHz
7. How many clock cycles are required to enter the data into the register in the following Figure?



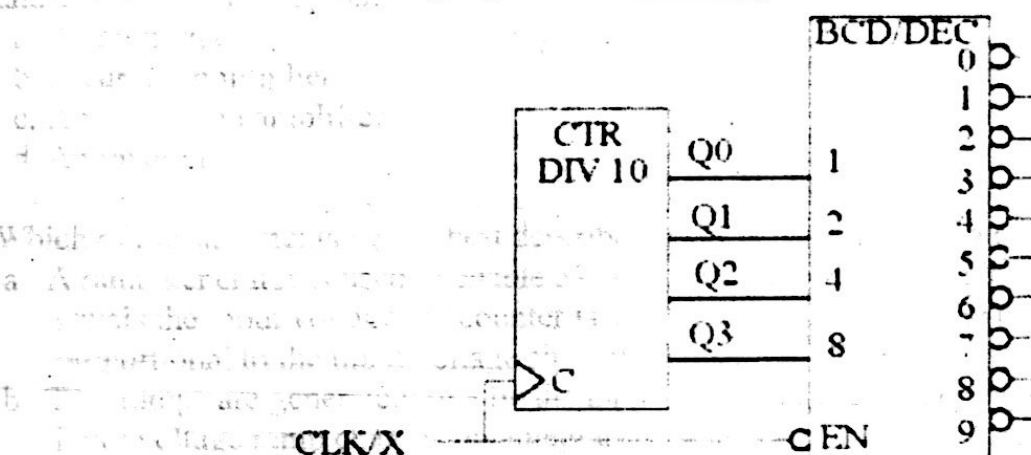
- a. 31 b. 32 c. 4 d. 5
8. What is the modulus of a 6-stage Johnson counter?
- a. 12 b. 64 c. 32 d. 6

9. During period D on the timing diagram shown below, the decimal count sequence will be _____.



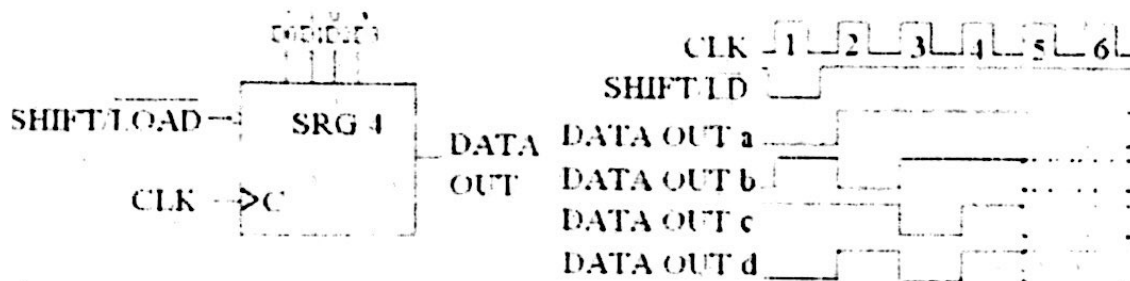
- a. 8, 9, 0, 1
- b. 0, 1, 9, 8
- c. 7, 8, 9, 1
- d. 1, 0, 9, 8

10. The purpose of the connection between 'C' of the 'CTR DIV 10' circuit and the 'EN' of the 'BCD/DEC' circuit in following Figure is to _____ and is referred to as _____.



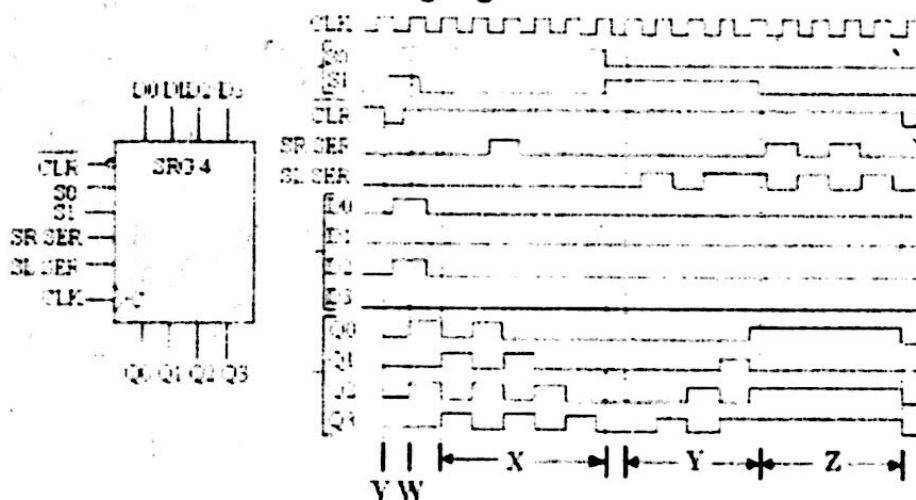
- a. speed up the counter, strobing
- b. clock the counter, strobing
- c. reduce the number of inputs, enabling
- d. eliminate glitches, strobing

11. Which of the DATA OUT waveforms in the following Figure is correct?



- DATA OUT d
- DATA OUT c.
- DATA OUT a.
- DATA OUT b.

12. What type of device is shown in the following Figure:



CLEAR	MODE		CLK	INPUTS		OUTPUTS			
				SERIAL		PARALLEL			
	SI	SO		LEFT	RIGHT	LD	DI	DO	DO
L	X	X	X	X	X	X	X	X	X
H	X	X	1	X	X	X	X	X	X
H	H	H	↑	X	X	a	b	c	d
H	L	H	↑	X	H	X	X	X	X
H	L	H	↑	X	L	X	X	X	X
H	H	L	↑	H	X	X	X	X	X
H	H	L	↑	L	X	X	X	X	X
H	L	L	X	X	X	X	X	X	X

Figure 1

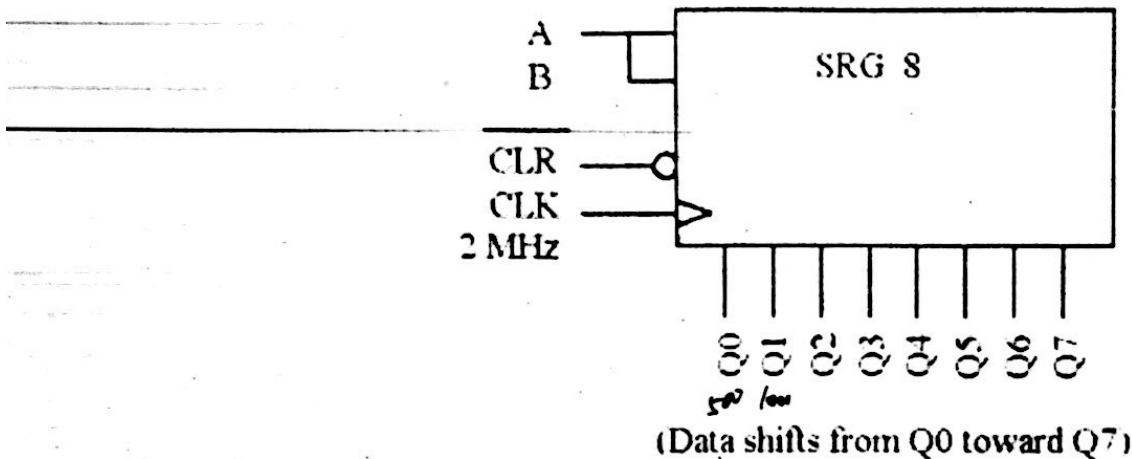
- 4-bit bidirectional universal shift register
- 2-way parallel-in serial-out bidirectional register
- Parallel-in parallel-out shift register with bidirectional data flow
- 2-bit serial-in 4-bit parallel-out bidirectional shift register

13. Referring to Figure 1, what action takes place during the interval labeled 'X' on the timing diagram?

- Data is shifted left through the register.
- Parallel data is entered into the register.
- Data is shifted right through the register.
- Serial data is entered into the register.

14. In order to use a shift register as a counter, _____
- serial-in serial-out register must be used.
 - the register's serial input is the counter input and the serial output is the counter output.
 - the serial output of the register is connected back to the serial input of the register.
 - the parallel inputs provide the input signal and the output signal is taken from the serial data output.

15. A serial data path needs a 2000 ns delay. Which output from the circuit below will provide the correct delay?



- a. Q7 b. Q3 c. Q1 d. Q0

$$8 \times 500 \times 10^{-9} = 4000 \text{ ns}$$

16. What type of op-amp circuit does not typically have a feedback path connected between its output and either of its input pins?

- A comparator
- A current multiplier
- An open-loop amplifier
- An inverter

17. Which of the statements below best describes the basic operation of a dual-slope A/D converter?

- A ramp generator is used to enable a counter through a comparator. When the ramp voltage equals the input voltage the counter is latched and then reset. The counter reading is proportional to the input voltage since the ramp is changing at a constant V/second rate.
- Two ramps are generated: one by the input voltage and the other by a reference voltage. The input voltage ramp charges the integrating capacitor, while the reference voltage discharges the capacitor and enables the counter until the capacitor is discharged, at which time the counter value is loaded into the output latches.
- The input voltage is used to set the frequency of a voltage-controlled oscillator (VCO). The VCO quits changing frequency when the input voltage stabilizes. The frequency of the VCO, which is proportional to the analog input voltage, is measured and is displayed on the digital display as a voltage reading.
- A ramp voltage and analog input voltage are applied to a comparator. As the input voltage causes the integrating capacitor to charge, it will at some point equal the ramp voltage. The ramp voltage is measured and displayed on the digital panel meter.

18. The 2's complement of binary 110110 is _____.

- a. 110100 b. 001001 c. 001010 d. 101010

19. What is the linearity of a D/A converter?

- The ability to resolve between forward and reverse steps when sequenced over its entire range of inputs.
- It is the reciprocal of the number of discrete steps in the D/A output.
- It is the comparison between the actual output of the converter and its expected output.
- It is the deviation between the ideal straight-line output and the actual output of the converter.

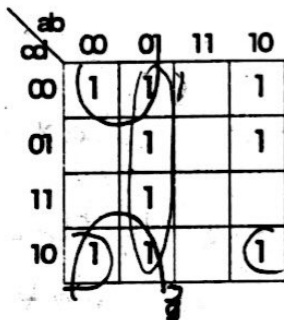
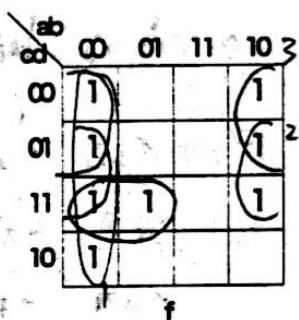
20. How many gates, including inverters, are required to implement the equation,

$$X = A\overline{C}D + AB(CD + BC), \text{ after simplification with Boolean algebra?}$$

- 3
- 5
- 7
- 9

Part III: SHORT ANSWER. Write the word, phrase or draw a block diagram that best completes each statement or answers the question.

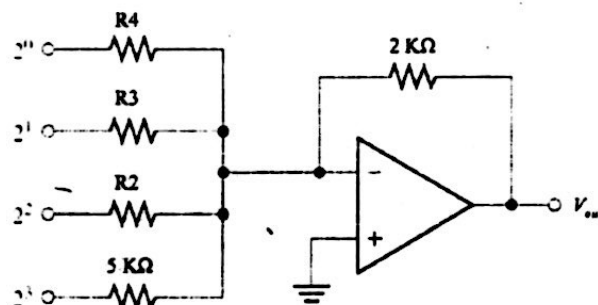
- There are 4 decoders and two OR gates. The inputs number of each OR gate is less than 8. Using the decoders and OR gates to implement the following two functions expressed in Karnaugh map. (Hint: implement the four min-terms in a same row/column with a single output of a decoder)



Decoder's truth table:

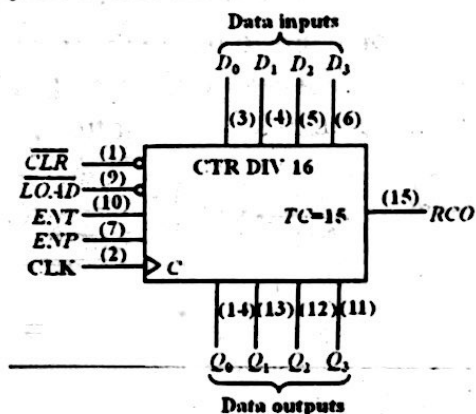
EN	a	b	0	1	2	3
0	x	x	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

2. For the Binary-Weighted-Input Digital-to Analog Converter (DAC) in the figure, please answer the questions below.
- (a) Please determine the ideal resistance values for the resistors $R_2 \sim R_4$ in the DAC in the following figure.
- (b) Now suppose the actual resistance values of $R_2 \sim R_4$ are 10% higher than the ideal values, respectively. Please determine the output voltage (in volts) and the conversion error (in percentage) as compared with an DAC with ideal $R_2 \sim R_4$ resistance values, when the digital input is $(1011)_2$ and input voltages are ideal (5V for HIGH and 0V for LOW).



3. Design a serial signal ($Y=1011100110$) generator by using one 74HC163 counter and a 1 of 8 data selector.

The logic symbol and truth table of 74HC163 are as following:

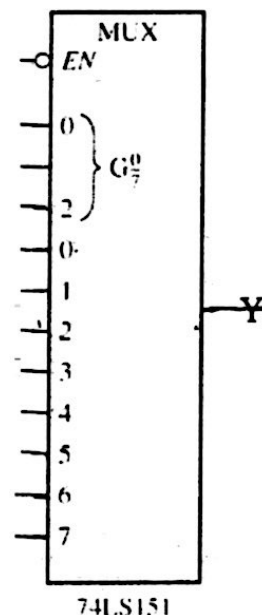
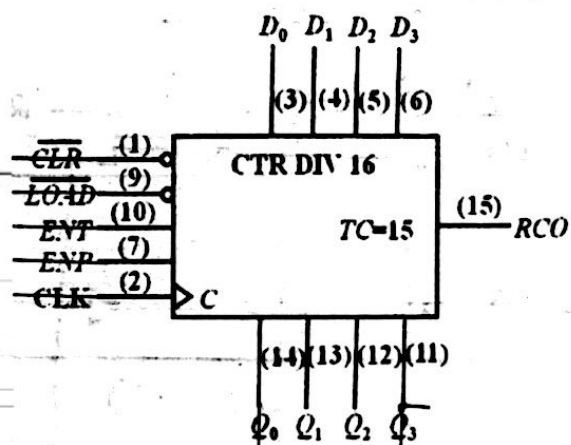


Operating Mode	$\overline{\text{CLR}}$	CLK	ENP	ENT	$\overline{\text{LOAD}}$	D	Q	RCO
Reset (Clear)	0	$\uparrow$	X	X	X	X	0	0
Parallel load	1	$\uparrow$	X	X	0	D	D	0
Count	1	$\uparrow$	1	1	1	X	Count	$\uparrow$
Inhibit	1	X	0	X	1	X	Q	$\uparrow$
	1	X	X	0	1	X	Q	0

† The RCO output is 1 when ENT is 1 and the counter is at TC (1111).

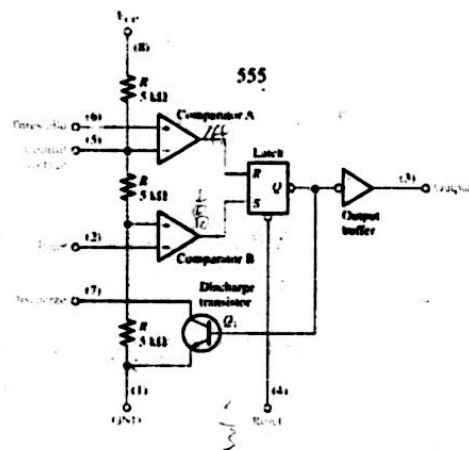
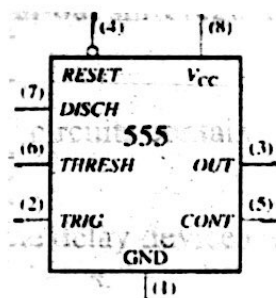
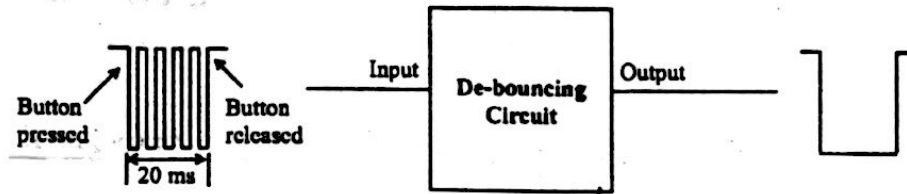
The High-to-Low transition of ENP and ENT should only occur while CLK is HIGH for conventional operation.

The Low-to-High transition of $\overline{\text{CLR}}$ and $\overline{\text{LOAD}}$ should only occur while CLK is HIGH for conventional operation.



74LS151

4. When a push button is pressed or released, the contacts will bounce between the *on* and *off* states for a fraction of a second. Because of this, the output from a push button must be *de-bounced* before it can be used by other digital circuits. The figure below illustrates the output from a button press. Please answer the following questions.
- Explain why and how a 555 timer can be used as a de-bouncing circuit;
 - Assume the resistance value of all resistors is $10K\Omega$. Draw the de-bouncing circuit with proper capacitor values in the 555 timer below.



5. Design a serial detector which can detect the 1101 appeared, by using two edge triggered J-K Flip Flops. The detector has one input and one output.

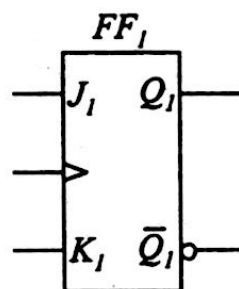
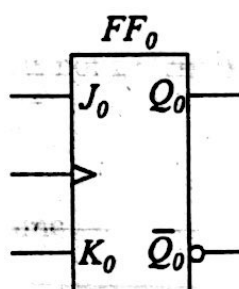
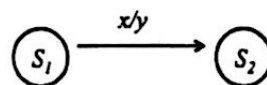
The following is an example of the relationship between the input and output.

Input x 0 0 1 1 0 1 1 0 1 0 1

Output y 0 0 0 0 0 1 0 0 1 0 0

- (a) Draw the state diagram of the detector.

(Hint: the state diagram has 4 states. And denote state transitions with input x and output y as the example shown on the right.)



上 海 交 通 大 学 试 卷 (B 卷)

(20 11 至 20 12 学 年 第 1 学 期)

班级号 _____ 学号 _____

课程名称 数字电子技术

成绩

1. (10%) Use the minimum number of gates to implement a circuit that detects if a 4-bit binary number $A_3A_2A_1A_0$ is greater than 9. When the number is greater than 9, the output $Y = 1$; otherwise $Y = 0$.

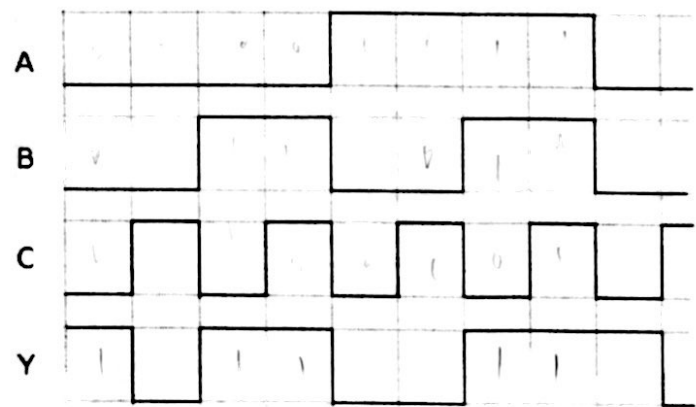
我承诺，我将严
格遵守考试纪律。
签名：

题号	1	2	3	4	5	6	7	8	9	
得分										
批阅人(流水阅卷)										

2. (10%) Find the minimum POS form for the following expression

$$Y = \overline{(A \oplus B) \overline{B} (\overline{A} \overline{B} \overline{C} \overline{D} + \overline{A} \overline{C} \overline{D})}$$

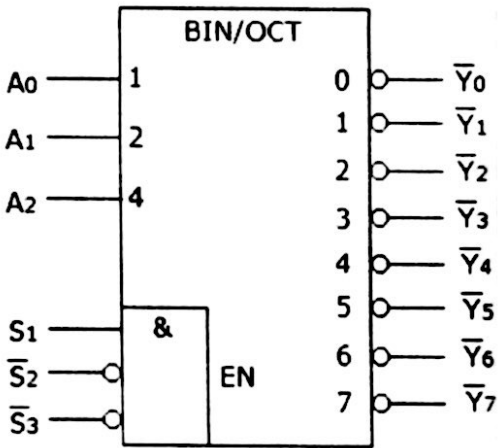
3. (15%) Use one 3-line-to-8-line decoder 74LS138 and one NAND gate to implement a circuit whose output Y and inputs A, B, and C are related as shown in Figure 1.



(a) Waveforms for the desired circuit

INPUTS					OUTPUTS							
S ₁	$\overline{S_2} + \overline{S_3}$	A ₂	A ₁	A ₀	$\overline{Y_0}$	$\overline{Y_1}$	$\overline{Y_2}$	$\overline{Y_3}$	$\overline{Y_4}$	$\overline{Y_5}$	$\overline{Y_6}$	$\overline{Y_7}$
0	x	x	x	x	1	1	1	1	1	1	1	1
x	1	x	x	x	1	1	1	1	1	1	1	1
1	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	1	0	1	1	0	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1	1	1
1	0	1	0	0	1	1	1	1	0	1	1	1
1	0	1	0	1	1	1	1	1	1	0	1	1
1	0	1	1	0	1	1	1	1	1	1	0	1
1	0	1	1	1	1	1	1	1	1	1	1	0

(b) Function Table for 74LS138



(c) Logic symbol for 74LS138

Figure 1

(10%) Use 4-bit adders to implement a circuit which adds up two 11-bit numbers. Label all critical inputs and outputs.

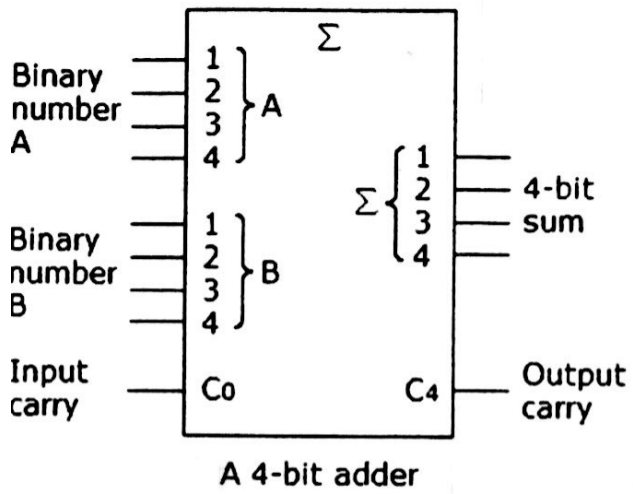


Figure 2

5. (15%) Use J-K flip-flops and the minimum number of gates to implement a synchronous 2-bit up/down counter with a Gray code sequence. The counter should count up when an $\overline{UP/DOWN}$ control input is 1 and count down when the control input is 0.

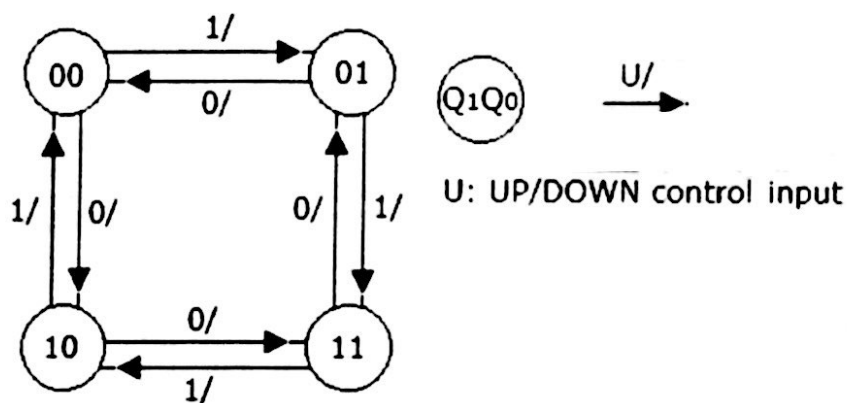


Figure 3

6. (15%) In Figure 4, $R_1 = 150 \text{ k}\Omega$, $R_2 = R_3 = 24 \text{ k}\Omega$, $C_1 = 0.033 \text{ }\mu\text{F}$, $C_2 = C_3 = C_4 = 0.01 \text{ }\mu\text{F}$. (1) What is the function of the circuit in block I? (2) What is the function of the circuit in block 2? (3) Suppose that the input v_{IN} , a single negative-going pulse, is applied at $t = 0$, sketch the waveforms v_{O1} , v_{O2} , and v_O . Suppose that both v_{O1} and v_{O2} are initially 0.

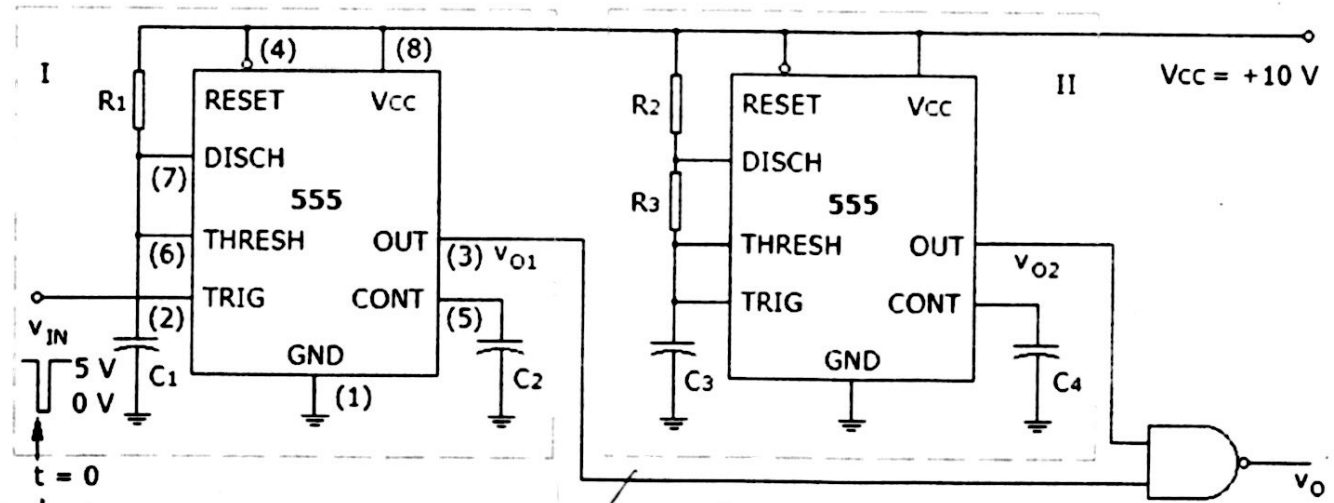
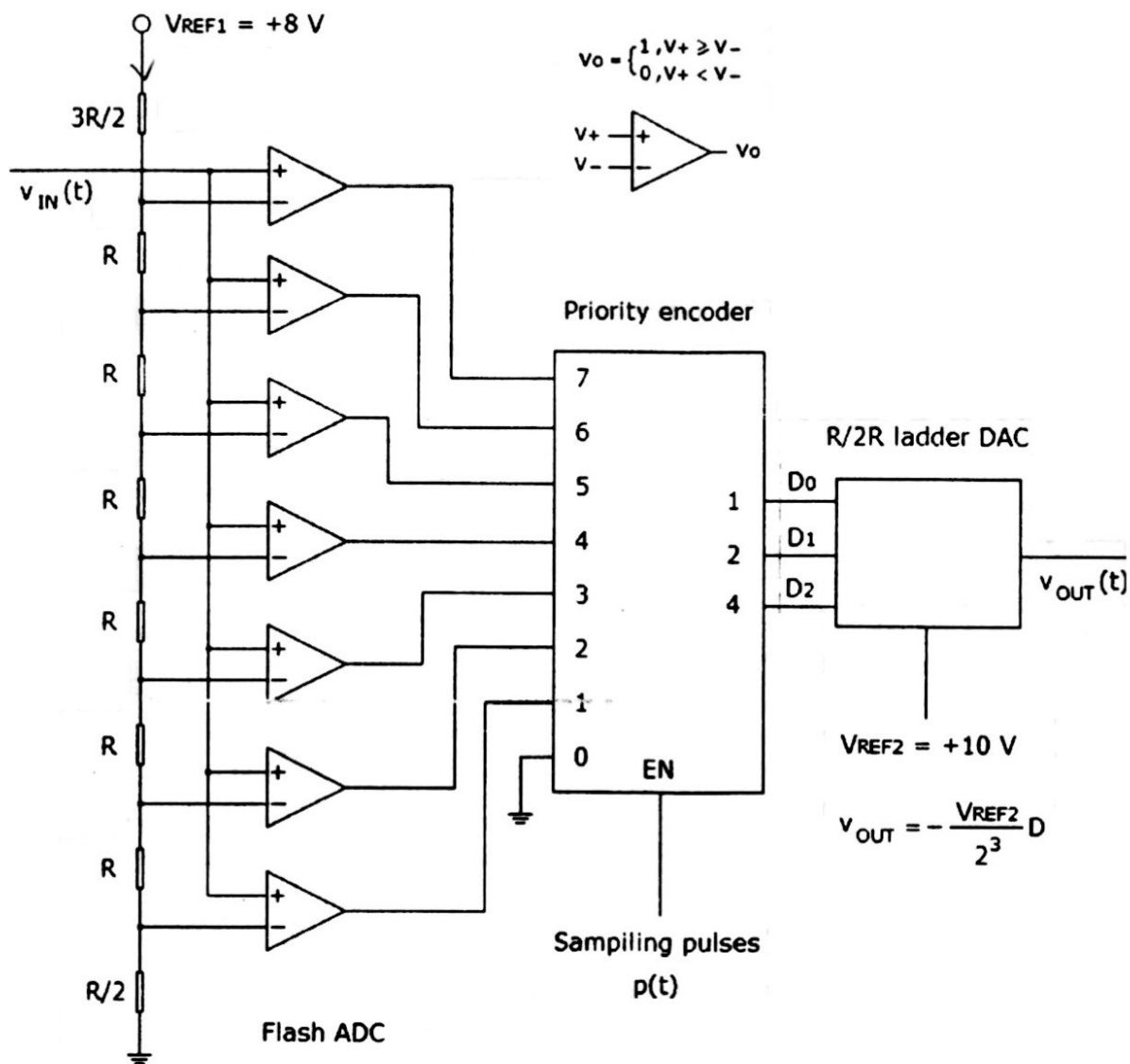


Figure 4

7. (15%) The input to the circuit in Figure 5 is $v_{IN}(t) = 4 + 4\sin(100\pi t)$ V, $0 \leq t \leq 20$ ms. The sampling signal has a frequency of 400 Hz. Determine the binary code sequence $D_2D_1D_0$ and the output $v_{OUT}(t)$ for $0 \leq t \leq 20$ ms.



8. (10%) Draw the state diagram for the circuit in Figure 6. Y is the output of the circuit.

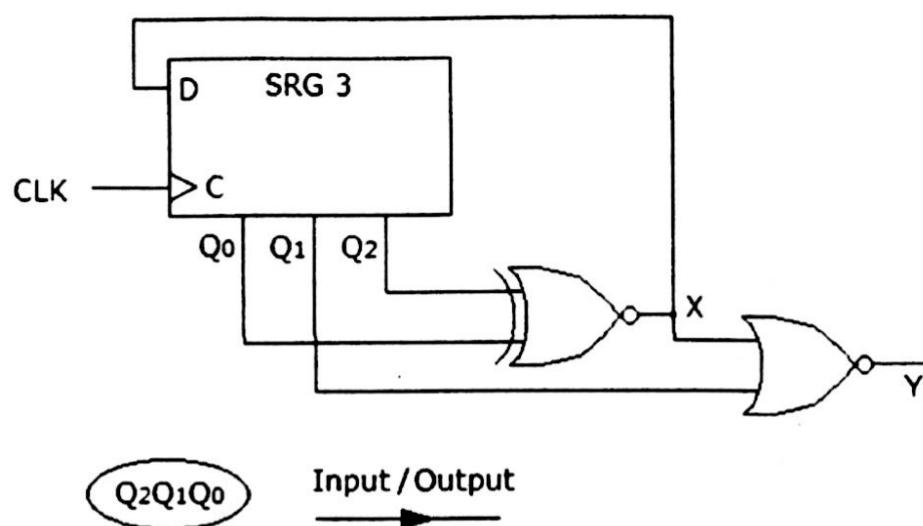


Figure 6

上 海 交 通 大 学 试 卷 (B 卷)

(2008 至 2009 学 年 第 1 学 期)

班级号 _____ 学号 _____ 姓名 _____
课程名称 _____ 数字电子技术 _____ 成绩 _____

PART I Multiple Choice (20%)

Answer the following questions 1-10 by filling out each parenthesis with a letter (A, B, C, or D), then transfer your answer to the following table.

Question	1	2	3	4	5	6	7	8	9	10
Answer										

1. The modulus of a counter is
 - (A) the number of flip-flops.
 - (B) the actual number of states in its sequence.
 - (C) the number of times it cycles in a second.
 - (D) the maximum possible number of states.

2. Which of the following codes exhibit even parity?
 - (A) 10011000.
 - (B) 01111000.
 - (C) 11010101.
 - (D) all.

3. What is the resolution of a 12-bit DAC?
 - (A) 8.33%.
 - (B) 0.0244%.
 - (C) 0.0488%.
 - (D) 0.083%.

4. CMOS operates more reliably than TTL in a high-noise environment because of its
 - (A) lower noise margin.
 - (B) input capacitance.
 - (C) high noise margin.
 - (D) smaller power dissipation.

我承诺，我将严格遵守考试纪律。

承诺人：_____

题号	I	11	12	13	14	15	16	17		
得分										
批阅人(流水阅卷教师签名处)										

5. Edge-triggered flip-flops must have
 - (A) very fast response times.
 - (B) at least two inputs to handle rising and falling edges.
 - (C) a pulse transition detector.
 - (D) active-low inputs and complemented outputs.
6. What advantage does a J-K flip-flop have over a R-S flip-flop?
 - (A) It has fewer gates.
 - (B) It has only one output.
 - (C) It has no invalid states.
 - (D) It does not require a clock input.
7. In an R/2R DAC, there are
 - (A) four values of resistors.
 - (B) one resistor value.
 - (C) two resistor values.
 - (D) a number of resistor values equal to the number of inputs.
8. Which of the following is a type of shift register counter?
 - (A) decade.
 - (B) binary.
 - (C) ring.
 - (D) BCD.
9. The storage cell in a SRAM is
 - (A) a latch.
 - (B) a capacitor.
 - (C) fuse.
 - (D) a magnetic domain.
10. Asynchronous counters are often called _____ counters.
 - (A) toggle.
 - (B) ripple.
 - (C) binary.
 - (D) flip-flop.

PART II Problems (80%)

11. A 555 timer and a nonretriggerable One-Shot 74121 are configured as shown in figure 1.

Determine the frequency of the output Q_1, Q_2 and the duty cycle respectively. And draw the waveform of the outputs Q_1, Q_2 . (15%)

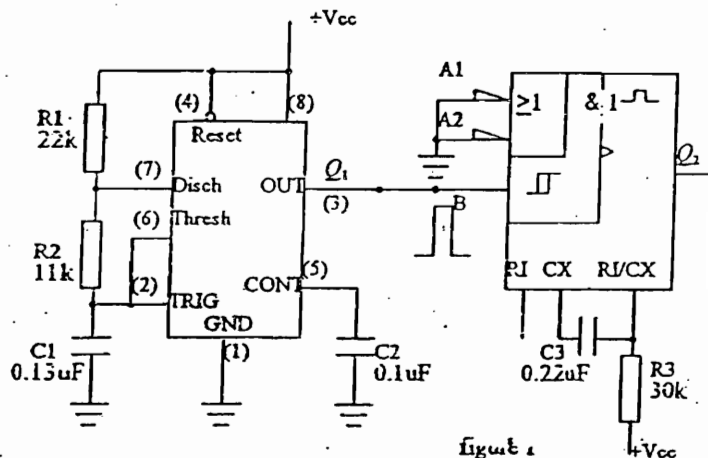
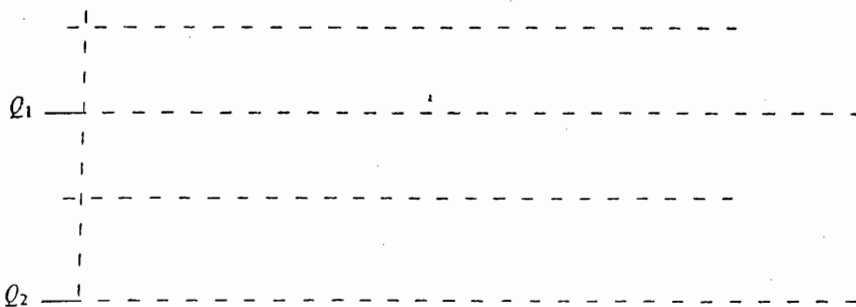
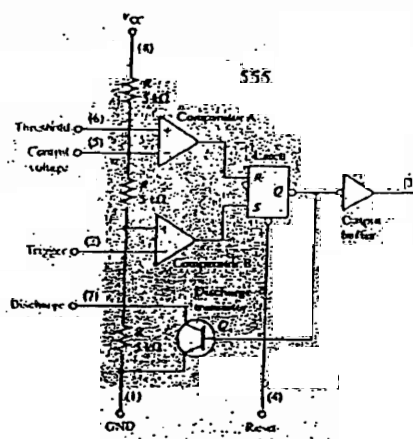


Figure 1



12. Design a shift register counter to produce the following binary sequence, Use D flip-flops.
(10%)

0, 1, 3, 6, 5, 2, 4, 0,...

13. Draw the outputs Q_1 and Q_2 of the master-slave J-K flip-flop for the input waveforms shown in figure 2. The flip-flop starts out RESET. (10%)

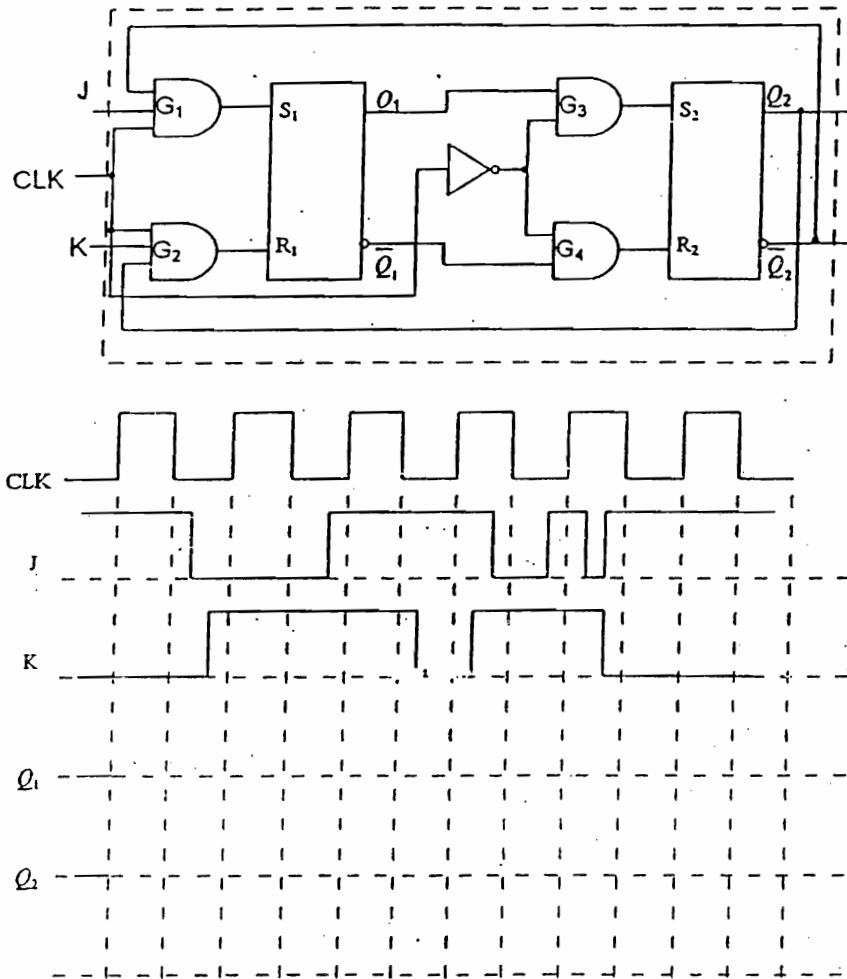


figure 2

14. Using Boolean algebra to convert the following expression to minimized SOP form.(10%)

$$F = \overline{\overline{AB} \cdot \overline{B+D} \cdot \overline{CD}} + BC + \overline{A} \cdot \overline{BD} + A + \overline{CD}$$

15. Design one half-adder by using TWO one-bit magnitude comparators (shown in figure 3) and inverters (10%)

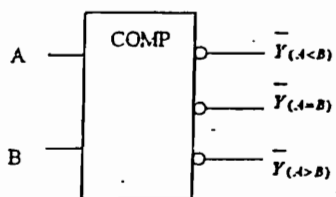
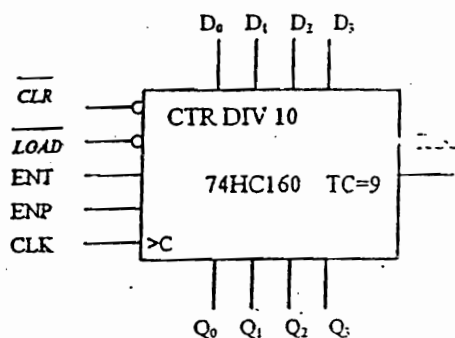


figure 3

16. Design a counter to produce the following binary sequence by using ONE BCD decade counter 74HC160 (shown in figure 4) and a few NAND gates. (15%)

0, 1, 2, 3, 5, 6, 7, 9, ...



74HC160 Function Table

CLK	$\overline{CLR}$	$\overline{LOAD}$	ENP	ENT	Function
X	0	X	X	X	Clear
↑	1	0	X	X	Preset
X	1	1	0	1	Inhibit
X	1	1	X	0	Inhibit($rc0=0$)
↑	1	1	1	1	Count

figure 4

17. Analyze the following sequential circuit (shown in figure 5), derive the logic expressions for flip-flop outputs Q_0 , Q_1 , and Y , and draw its state diagrams. (10%)

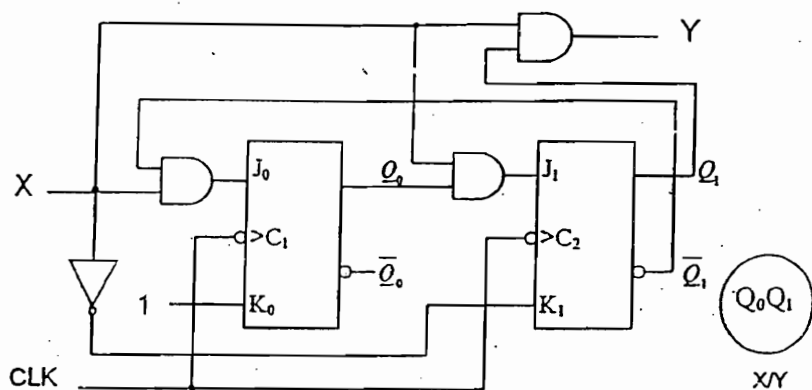


figure 5

上海交通大学试卷(A卷)

(20 07 至 20 08 学年 第 1 学期)

班级号 _____ 学号 _____ 姓名 _____
 课程名称 数字电子技术 成绩 _____

PART I Multiple Choice (20%)

Answer questions 1-10 by filling out each parenthesis with a letter (A, B, C, or D), then transfer your answer to the following table.

Question	1	2	3	4	5	6	7	8	9	10
Answer										

- If a NAND gate is being used in a circuit, any of its unused inputs should be tied (), or to another input on the same gate.
 A. LOW B. 0 C. to the ground D. HIGH
- The expression $\overline{A \cdot B} = \overline{A} + \overline{B}$ is called ().
 A. DeMorgan's theorem B. absorption property C. consensus theorem D. distributive law
- Four types of devices that are classified as PLDs are PROM, PLA, PAL, and GAL. The () is a PLD that consists of a programmable AND array and a programmable OR array.
 A. PROM B. PLA C. PAL D. GAL
- Write the Boolean expression implemented by the PLD shown in Figure 1.

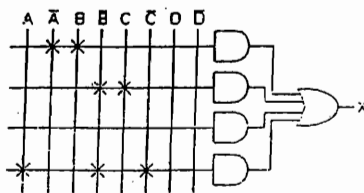


Figure 1

- $X = \overline{A}B + B\overline{C} + \overline{A}BC$
- $X = \overline{A}B + \overline{B}C + A\overline{B}C$
- $X = AB + BC + ABC$

10. Perhaps the fastest A/D converter is () ADC.

A. flash B. single-slope C. dual-slope D. successive-approximation

PART II (80%)

11. (10%) Using Boolean algebra, convert the following expression to minimized SOP form.

$$Y = ABC + ABD + \overline{A}BC + CD + B\overline{D} + ABCE.$$

12. (10%) Determine the truth table and Boolean expression for the circuit in Figure 4.

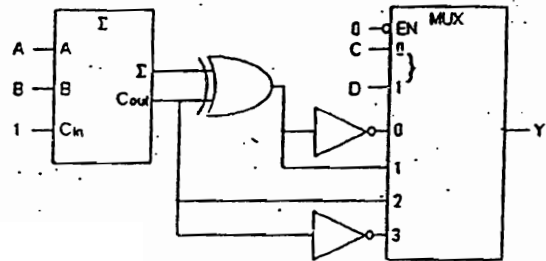


Figure 4

13. (10%) Design a combinational circuit that has two input signals X_0 and X_1 , two control signals C_0 and C_1 , and one output function F . The control signals have the following effect on the output:

C_0	C_1	F
0	0	0
0	1	X_0
1	0	$X_0 X_1$
1	1	$X_0 \oplus X_1$

Implement your logic by using one 74HC154 (1-of-16 decoder, shown in Figure 5) and one NAND gate.

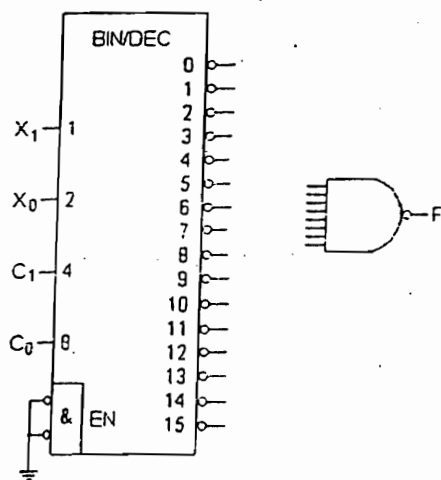


Figure 5

14. (10%) Determine the values of the external resistors for a 555 timer (shown in Figure 9) used as an astable multivibrator with an output frequency of 15kHz, if the external capacitor C is 0.001μF and the duty cycle is to be approximately 33.3%.

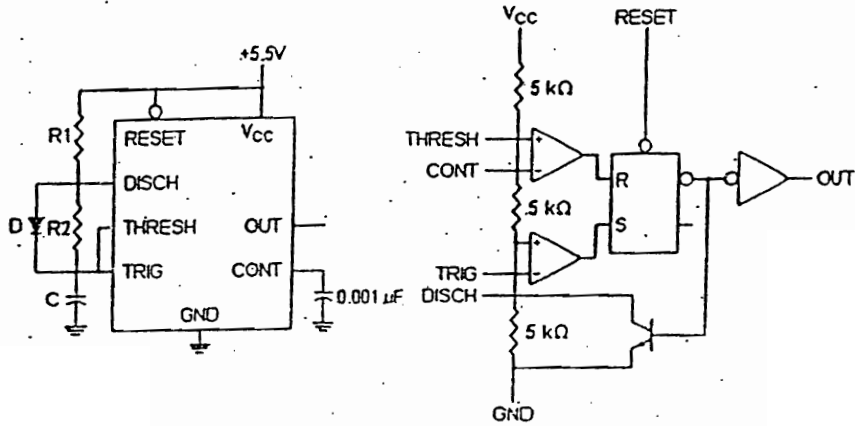


Figure 6

15. (10%) Use two IC 4-bit synchronous binary counters (shown in Figure 10) and one NAND gate to implement a divide-by-251 circuit. The function table for the IC counter is shown below.

Mode	CLK	CTEN	LOAD	$D_3 D_2 D_1 D_0$	$Q_3 Q_2 Q_1 Q_0$	TC
Parallel load	↑	X	L	$d_3 d_2 d_1 d_0$	$q_3 q_2 q_1 q_0$	L
Count	↑	H	H	X X X X	Count	H if $Q_3 Q_2 Q_1 Q_0 = HHHH$
Inhibit	X	L	H	X X X X	$Q_3 Q_2 Q_1 Q_0$	L

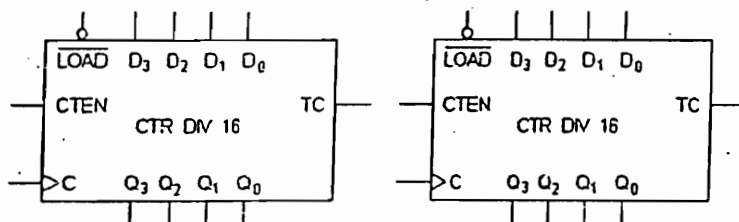


Figure 7

16. (10%) Use JK flip-flops and gates to implement a 2421 BCD counter. The 2421 code

$Q_3Q_2Q_1Q_0$ is shown in the table below.

Decimal	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	1	0	0	0
3	1	0	0	1
4	1	0	1	0
5	1	0	1	1
6	1	1	0	0
7	1	1	0	1
8	1	1	1	0
9	1	1	1	1

17. (10%) Expand the 256×4 ROM (shown in Figure 8) to form a 512×8 ROM.

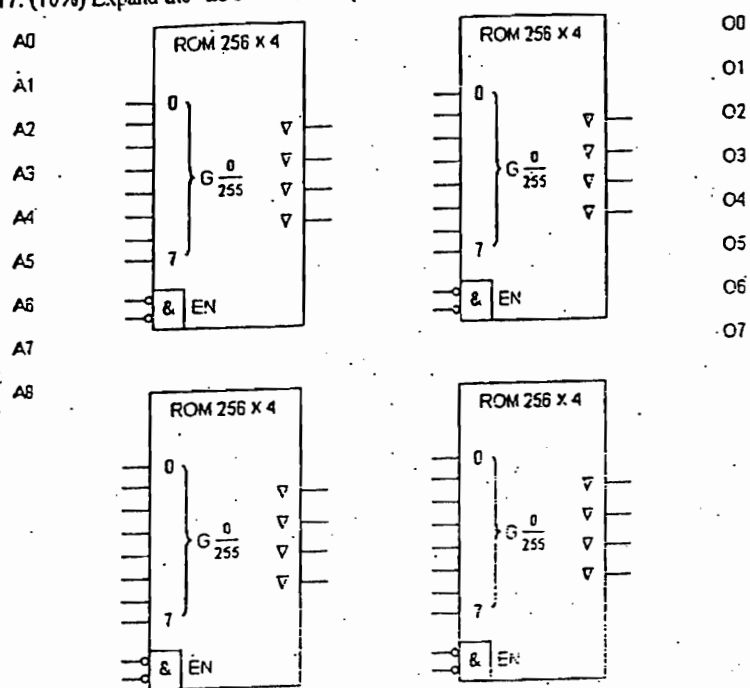


Figure 8

18. (10%) Sketch and label the output of the circuit in Figure 9. Suppose the initial state of the shift register is $Q_3Q_2Q_1Q_0 = 0010$.

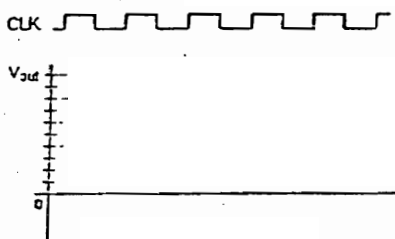
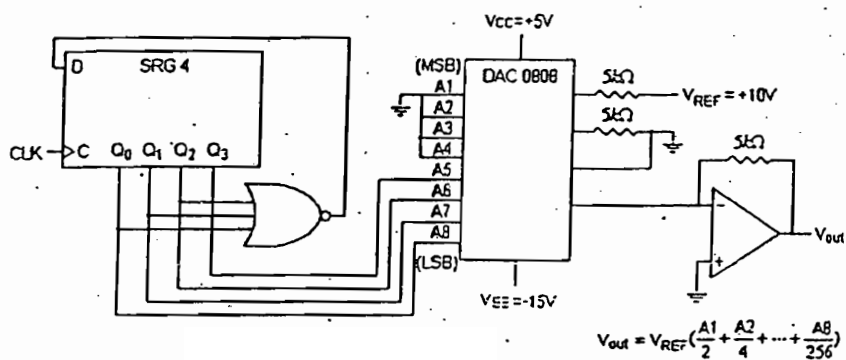


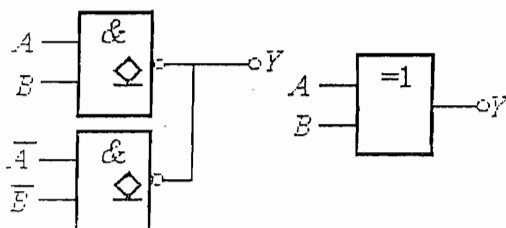
Figure 9

参考试卷 (5)

一、是非题 (对则打“√”, 错打“×”。每题 2 分, 共 10 分。)

() 1. 存储容量为 32KBit 表示存储器有 32000 个存储单元。

() 2. 下图中两电路的逻辑功能相同:



() 3. 4 位同步二进制加法计数器与 4 位异步二进制加法计数器的状态转换表不同。

() 4. 已知 $AB + C = AB + D$, 则可得 $C = D$ 。

() 5. 模 N 计数器可用作 N 分频器。

二、填空题 (每空 1 分, 共 20 分)

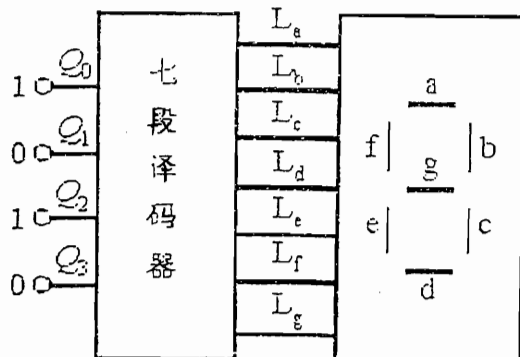
1. 数码 10000111 作为自然二进制数时相应的十进制数为 _____; 作为 8421 码时相应的十进制数为 _____, 该十进制数对应的自然二进制数为 _____。

2. 逻辑函数的 _____ 表达式是唯一的。

3. 门电路输出为 _____ 电平时的负载为拉电流负载, 输出为 _____ 电平时的负载为灌电流负载。

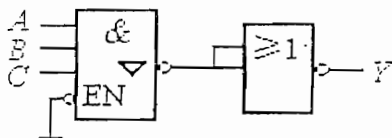
4. 右图所示的译码显示电路中, 输入为 8421BCD 码。设控制显示信号高电平有效, 则

$L_a L_b L_c L_d L_e L_f L_g =$ _____。



5. 一个4位移位寄存器, 经过 _____ 个时钟脉冲 CP 后, 4位串行输入数码全部存入寄存器; 再经过 _____ 个时钟脉冲 CP 后可串行输出4位数码。
6. 要组成模15计数器, 需要采用 _____ 个触发器。
7. 要用 n 位二进制数为 N 个对象编码, 必须满足 _____。
8. 2^n 选1数据选择器有 _____ 位地址码。
9. 存储容量为 1024×4 位 RAM, 其地址线有 _____ 条。
10. 当输入模拟电压最大值为 $+5V$ 时, 若采用8位 ADC, 则其分辨率为 _____。

11. 使右图中输出为0的输入变量的取值组合有 _____ 种。

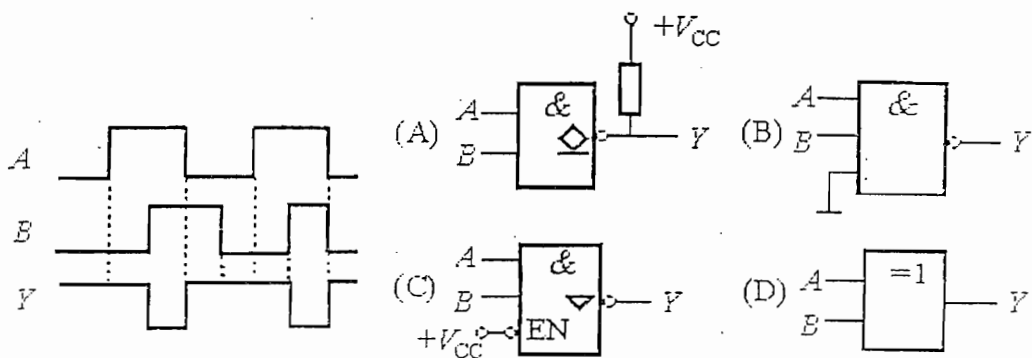


12. 设 A 表示1996年, B_1 、 B_2 分别表示6月和2月, C 表示1号, 则 $A \cdot B_1 \cdot C$ 表示 _____; $A \cdot \overline{B_1} \cdot \overline{B_2} \cdot C$ 有 _____ 天。
13. $Y = \underbrace{0 \oplus 0 \oplus \dots \oplus 0}_{1995 \text{ 个 } 0} = \underline{\hspace{2cm}}$ 。
14. A/D 转换一般分为 _____ 和 _____ 两步完成。

三、单项选择题 (每小题 2 分, 共 10 分)

- () 1. 已知逻辑函数 $Y = \overline{ABC + CD}$, 可以肯定 $Y = 0$ 的是
 (A) $A = 0, BC = 1$; (B) $BC = 1, D = 1$;
 (C) $AB = 1, CD = 0$; (D) $C = 1, D = 0$ 。

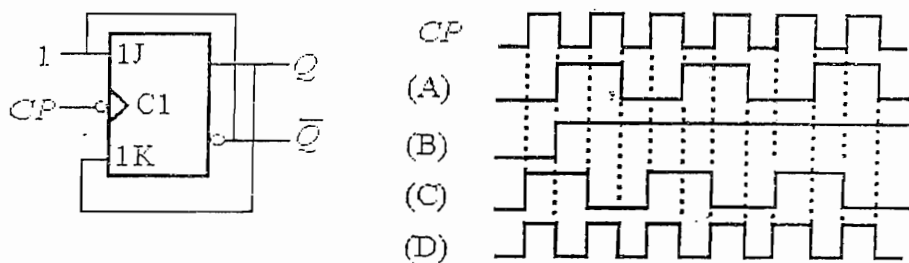
() 2. 输入 A 、 B 和输出 Y 的波形如图所示, 则对应的逻辑门电路是



() 3. 用来鉴别脉冲信号幅度时, 应采用

(A) 稳态触发器; (B) 双稳态触发器; (C) 多谐振荡器; (D) 施密特触发器。

() 4. 下图中, 正确的输出波形为



() 5. 构成同步二进制加法计数器时, 下列哪种做法是错误的

(A) 将各触发器接成 T 触发器;

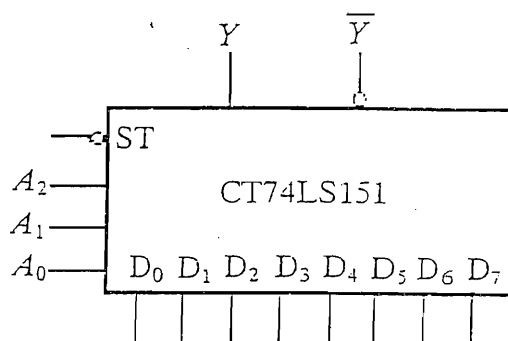
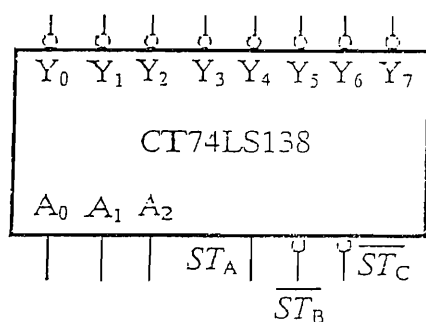
(B) $T_0 = 1$;

(C) $T_i = Q_{i-1} Q_{i-2} \cdots Q_0$, ($i = 1, 2, \cdots$);

(D) $CP_i = Q_{i-1}$, ($i = 1, 2, \cdots$).

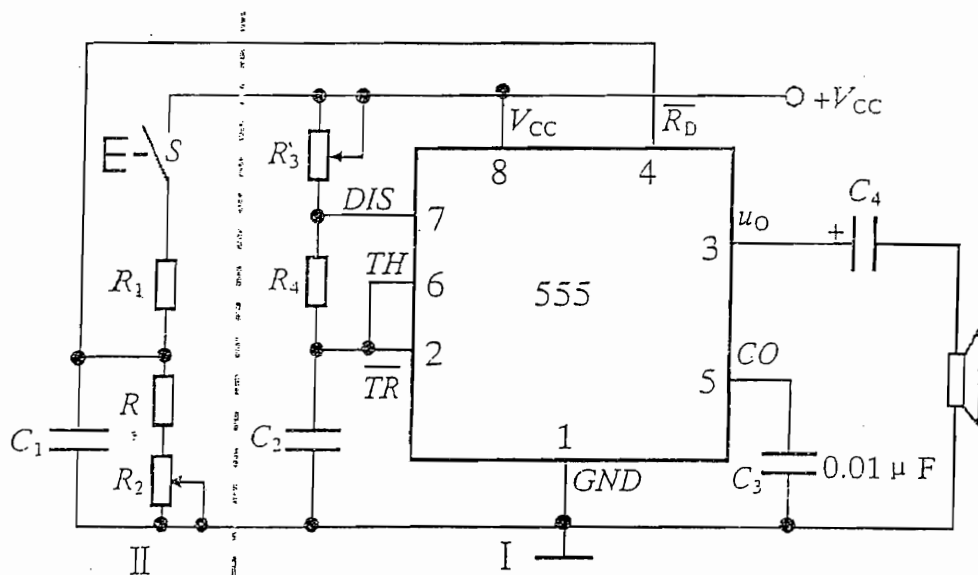
四、某工厂有一台容量为 35KW 的自备电源，为 A、B、C 三台用电设备供电，设 A、B、C 设备的额定功率分别为 10KW、20KW、30KW，它们投入运行是随机的组合。试分别用下面三种方案设计一个电源过载报警电路。

- (1) 用最简于非门电路实现之。
- (2) 用数据选择器 CT74LS151 实现之。
- (3) 用 CT74LS138 和门电路实现之。 (20 分)



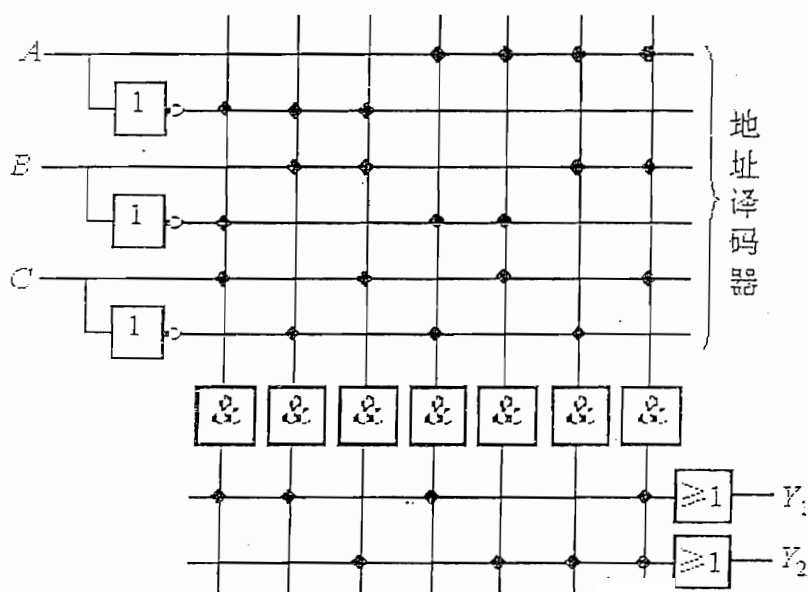
五、下图为简易门铃电路，设电路中元器件参数合适， $R \gg R_1$ ， S 为门铃按钮，当按钮按一下放开后，门铃可响一段时间。

- (1) 指出电路“Ⅰ”的名称。
- (2) 分析电路“Ⅱ”在门铃电路中的作用。
- (3) 欲调高铃声的音调，应如何调节？
- (4) 欲延长门铃响的时间，又如何调节？ (10 分)



六、试分别用计数状态 0000 ~ 1011 和 0100 ~ 1111 实现十二进制计数器 (可选用的集成计数器为 CT74LS290、CT74LS160 ~ 163)。(15 分)

七、试分析下图所示 PROM 电路的逻辑功能。要求写出标准与或式，列出真值表，并说明逻辑功能。(10 分)



八、卡诺图法化简。(5 分)

已知 $X = AB + \bar{A}C$, $Y(A, B, C) = \sum m(0, 1, 3, 4, 5, 6, 7)$

求 $F(A, B, C) = X + \bar{Y}$ 的最简与或式。

参考试卷 (5) 解答

一、是非题 (10 分)

1. × 2. × 3. × 4. × 5. √

二、填空题 (每空 1 分, 共 20 分)

1. 135, 87, 1010111; 2. 标准与或; 3. 高, 低;
 4. 1011011; 5. 4, 4; 6. 4; 7. $2^n \geq N$;
 8. n ; 9. 10; 10. 19.53mV; 11. 7;
 12. 1996 年 6 月 1 日, 10; 13. 0;
 14. 采样保持, 量化编码。

三、单项选择题 (每小题 2 分, 共 10 分)

1. B; 2. A; 3. D; 4. A; 5. D。

四、(20 分)

解: 设设备 A、B、C 运行为 1, 否则为 0; 报警信号用 Y 表示, 报警为 1, 否则为 0。则可得真值表为

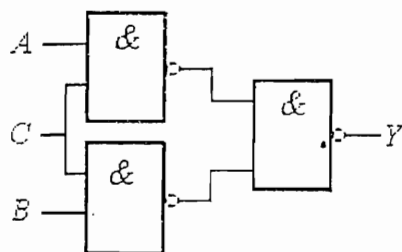
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

(5 分)

方案(1):

BC	00	01	11	10
A				
0			1	
1		1	1	

$$Y = BC + AC = \overline{\overline{BC} \cdot \overline{AC}}$$



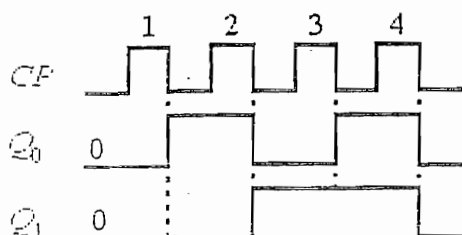
(5 分)

八、(5 分)

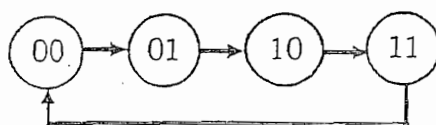
解: $u_O = u_I$

九、(8 分)

解: 异步 2 位二进制加法计数器。 (4 分)



(2 分)



(2 分)

十、(12 分)

解: 有多种方案, 参考答案之一为

CP	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1/0	1/0	1/0

